



Report

Robot Navigation with the hardware

Dinh-Son Vu

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1. Pre-requisites

- Ubuntu 24.04 Noble can run on your computer (VM, UTM, or dual boot).
- ROS2 Jazzy is installed.
- This is NOT a simulation. Gazebo is no longer used. The hardware is required.
- VSCode and its extension PlatformIO (PIO) should be installed.
- The esp32 and the lidar should have been tested while being connected on the RPI. For this package, the robot should NOT be connected to the computer.

2. Tasks

There are no packages to be installed, but make sure that the PC and RPI launch the right file respectively:

- PC should launch: rviz, slam, nav2
- RPI should launch: controller, rplidar

The bringup file for each platform should be modified to launch the correct packages.

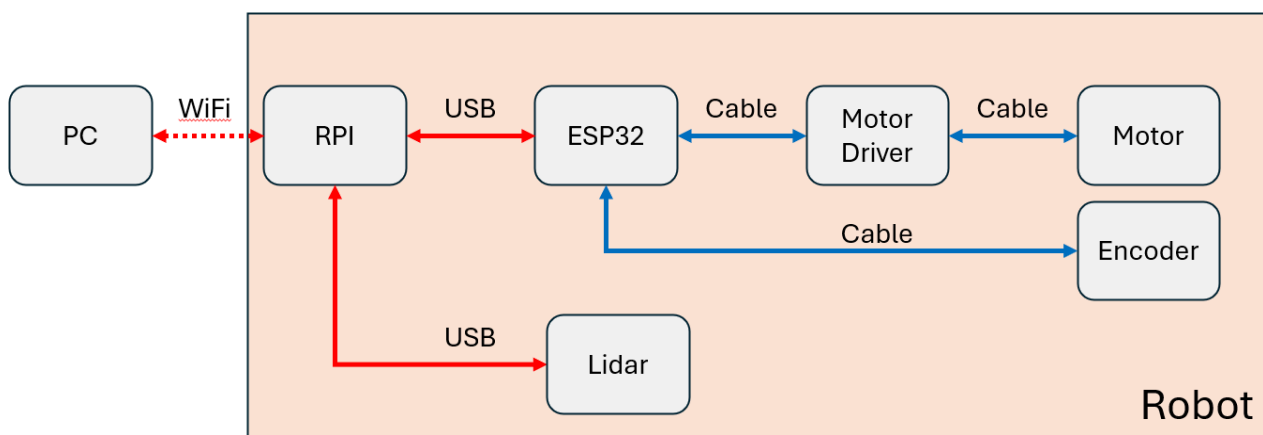


Figure 1: Robot configuration and connection

3. Explanation

One of the benefits of ROS2 is its abstraction regarding communication: as long as the RPI and PC are on the same network, they share the topics and nodes.

4. Packages, nodes, and topics

Packages	Nodes	Description
rmitbot_bringup		Launch the other packages and nodes
rmitbot_description	display.launch.py: rviz, rsp gazebo.launch.py: gazebo	Launch the urdf and static tf
rmitbot_controller	Controller manager: jsb spawner, controller spawner	Launch input/output of the robot
rmitbot_localization	ekf	Launch the sensor fusion between wheel encoder and imu
rmitbot_mapping	Slam rplidar	Launch the lidar and slam
rmitbot_navigation	nav2	Launch the navigation
rmitbot_vision	apriltag	Include the plugin for camera. Launch the apriltag detection
rmitbot_firmware	Arduino firmware Hardware interface	The arduino firmware is the code flashed on the esp32. The hardware interface is the node launched with ROS2.

5. Future works

The robot may require fine tuning, as the simulation and the real hardware are rarely perfectly matching.